



Discovering Directions of Uncertainty in Speech Inpainting

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Abstract

Speech inpainting aims to restore missing segments in audio signals. While both classical signal processing and deep learning approaches have been developed for this task, they typically generate a single output, despite the existence of multiple plausible reconstructions. In this paper, we adapt Neural Posterior Principal Component (NPPC) to recover the principal directions of variation in the posterior distribution of the original signal given its masked version. Our empirical results demonstrate that these directions capture diverse and meaningful variations in both speech content and style, while more precisely capturing the predictive error compared to a more costly Bayesian deep learning approach.

Index Terms: Speech Inpainting, Uncertainty Quantification

🔗 <https://github.com/kfirc1503/uq-inpaint>

1. Introduction

Audio signals often contain corrupted segments where multiple samples are missing, noisy, or unreliable. These distortions can arise from various factors, including recording errors, packet loss during transmission, or the degradation of storage media over time. Audio inpainting aims to reconstruct these missing or corrupted portions, restoring the signal's integrity.

Early approaches to audio inpainting primarily relied on signal processing techniques and sparse representations. Adler et al. [1] introduced a foundational audio inpainting framework, leveraging dictionary learning methods in the time-frequency domain. Mokry et al. [2] proposed a probabilistic framework based on nonnegative matrix factorization. They treat the missing samples as latent variables, and derive expectation-maximization algorithms for estimating the model parameters in either the time or frequency domains. Another traditional approach is based on autoregression. The Janssen algorithm [3], widely regarded as the state-of-the-art method for time-domain audio inpainting, has been effective for reconstructing gaps up to 50 ms in length. Recent work by Mokry et al. [4] extends this approach to the time-frequency domain, potentially enhancing its applicability to longer gaps.

The rise of deep learning has introduced new possibilities for audio inpainting. Ebner et al. [5] explored the use of generative adversarial networks (GANs) for audio inpainting, showing promise in generating longer segments of missing audio. Federico et al. [6] proposed using multi-resolution harmonic CNNs for audio inpainting. Recent works have introduced diffusion-based audio inpainting [7, 8], capable of handling gaps of various lengths. Another trend is to incorporate visual information, leveraging accompanying video data to guide the audio inpainting process [9, 10].

In many cases, especially for speech gaps of several tens of milliseconds, a single definitive reconstruction may not exist; instead, multiple plausible completions can be considered valid. However, previous studies have largely overlooked this uncertainty.

Uncertainty quantification aims to assess model confidence. Techniques such as Monte Carlo (MC) Dropout [11], deep ensembles [12], and test-time augmentations [13] estimate prediction variability. Recently, quantile regression and conformal prediction [14] gained attention for offering distribution-free uncertainty bounds. Predictive uncertainty in audio tasks remains relatively underexplored. Recent work has begun addressing this gap in areas such as speech enhancement [15] and source localization [16, 17, 18, 19].

In this paper, we introduce a framework to quantify and track uncertainty in speech inpainting. Building on [20], we leverage NPPC to recover the principal directions of variation in the posterior distribution of the clean signal given the distorted input. Our experimental results show that these directions capture diverse variations in both speech content and style, manifested in hearing, transcription, and pitch. Moreover, compared to a Bayesian deep learning approach, NPPC can better reconstruct predictive error with reduced computational cost. An illustration of our approach is provided in Fig. 1.

2. Preliminaries

Consider a clean speech signal represented by its short time Fourier transform (STFT) $\mathbf{X} \in \mathbb{C}^{F \times T}$, where F denotes the number of frequency bins and T represents the number of time frames. We observe a masked version of the signal, $\mathbf{X}_m$, where certain time frames are missing or severely corrupted. Let $\mathbf{m} \in [0, 1]^T$ be a binary masking vector, where ones indicate observed frames and zeros denote distorted ones. The masked signal is given by

$$\mathbf{X}_m = \mathbf{X} \odot \mathbf{M}, \quad (1)$$

where $\odot$ denotes element-wise product, $\mathbf{M} = \mathbf{1}_F \cdot \mathbf{m}^T$, and $\mathbf{1}_F$ is an all-ones vector of length F . Note that the location of the masked part is assumed to be known.

Next, let $\mathbf{x}, \mathbf{x}_m \in \mathbb{R}^{2FT}$ denote the vectorized STFT representation, comprising the real and imaginary parts of $\mathbf{X}$ and $\mathbf{X}_m$, respectively, for all time frames and frequency bins. Given a dataset $\mathcal{D} = \{(\mathbf{x}_m, \mathbf{x})\}_N$, a model $f(\mathbf{x}; \theta)$ with parameters θ is trained to reconstruct the missing portions of $\mathbf{x}_m$. The reconstruction error is defined as

$$\mathbf{e}(\mathbf{x}_m, \mathbf{x}; \theta) = |\mathbf{x} - \hat{\mathbf{x}}|$$

where $\hat{\mathbf{x}} = f(\mathbf{x}_m; \theta)$ denotes the predicted reconstruction.

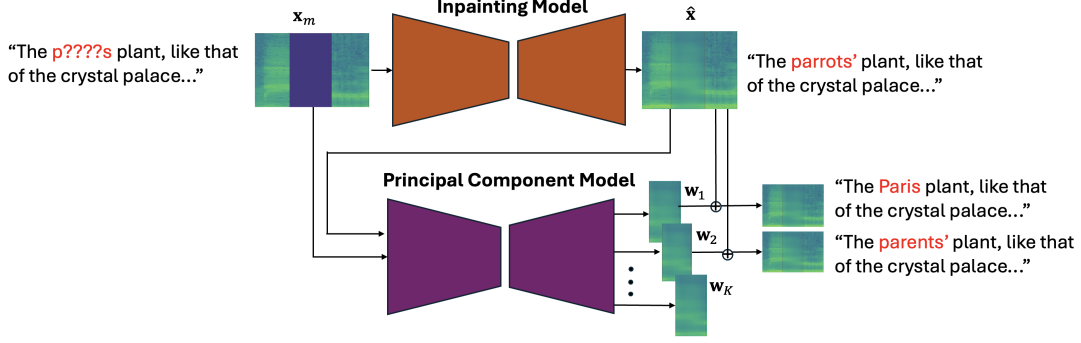


Figure 1: Illustration of our uncertainty quantification framework for speech inpainting using NPPC. A U-Net-based inpainting model generates a single reconstructed output for a given masked speech signal. To capture uncertainty, a second model learns the Principal Components (PCs) of the posterior probability distribution. By traversing along these directions, we generate diverse output possibilities with varied transcriptions, reflecting the inherent uncertainty in speech inpainting task.

Table 1: Layerwise summary of U-Net networks for the inpainting model and PC model. The models are the same except for the first and last layers. Whenever there is a difference, we use inpainting/PC channels notation. k denotes the kernel size.

Layer	Composition	In Chan.	Out Chan.	Hyperparameters
1	Input Convolution (Double Conv)	1/2	64	$k = 3, \text{pad} = 1$
2	Downsample Block 1 (MaxPool + Double Conv)	64	128	$\text{pool} = 2, k = 3, \text{pad} = 1$
3	Downsample Block 2 (MaxPool + Double Conv)	128	256	$\text{pool} = 2, k = 3, \text{pad} = 1$
4	Downsample Block 3 (MaxPool + Double Conv)	256	512	$\text{pool} = 2, k = 3, \text{pad} = 1$
5	Downsample Block 4 (MaxPool + Double Conv)	512	512	$\text{pool} = 2, k = 3, \text{pad} = 1$
6	Upsample Block 1 (Upsample + Double Conv)	1024	256	$\text{scale} = 2, k = 3, \text{pad} = 1$
7	Upsample Block 2 (Upsample + Double Conv)	512	128	$\text{scale} = 2, k = 3, \text{pad} = 1$
8	Upsample Block 3 (Upsample + Double Conv)	256	64	$\text{scale} = 2, k = 3, \text{pad} = 1$
9	Upsample Block 4 (Upsample + Double Conv)	128	64	$\text{scale} = 2, k = 3, \text{pad} = 1$
10	Output Convolution (1x1 Conv)	64	1/5	$k = 1$

3. Method

Our goal is to quantify the uncertainty in inpainting outcomes by generating a diverse set of plausible alternatives rather than a single estimate, $\hat{x}$. This uncertainty is encoded in the posterior distribution of $\mathbf{x}$ given $\mathbf{x}_m$. We aim to identify multiple directions of variation that capture the principal modes of this posterior distribution. However, a single estimate, $\hat{x}$, conveys only the posterior mean, neglecting higher-order moments.

To solve this, we adopt the NPPC method [20]. Rather than explicitly modeling the entire posterior distribution, this approach directly learns the posterior PCs. To this end, we train a model $\mathbf{d}(\mathbf{x}_m, \hat{x}; \phi)$, parameterized by ϕ , to output the K principle directions of variation $\{\mathbf{d}_k\}_{k=1}^K$. These are transformed to the set of K PCs $\{\mathbf{w}_k\}_{k=1}^K$, by a Gram-Schmidt procedure, making the directions satisfy the orthogonality and unit-norm constraints by construction. Specifically, given a dataset $\mathcal{D}' = \{(\mathbf{x}_m, \mathbf{x}, \hat{x})\}_N$, the PCs are learned by maximizing their alignment with the reconstruction error $\mathbf{e}$, which is equivalent to minimizing the following loss function:

$$\mathcal{L}_w = - \sum_{(\mathbf{x}_m, \mathbf{x}, \hat{x}) \in \mathcal{D}'} \sum_{k=1}^K |\mathbf{w}_k(\mathbf{x}_m, \hat{x}; \phi)^T \mathbf{e}(\mathbf{x}_m, \mathbf{x}; \theta)|^2, \quad (2)$$

where $\{\mathbf{w}_k\}_{k=1}^K$ are computed by the following Gram-Schmidt procedure:

$$\mathbf{w}_1 = \frac{\mathbf{d}_1}{\|\mathbf{d}_1\|} \quad (3)$$

$$\mathbf{w}_k = \frac{\mathbf{d}_k - \sum_{l=1}^{k-1} (\mathbf{d}_k^T [\mathbf{w}_l]_{\text{sg}}) [\mathbf{w}_l]_{\text{sg}}}{\|\mathbf{d}_k - \sum_{l=1}^{k-1} (\mathbf{d}_k^T [\mathbf{w}_l]_{\text{sg}}) [\mathbf{w}_l]_{\text{sg}}\|}. \quad (4)$$

Here, $[\cdot]_{\text{sg}}$ denotes the StopGrad operation. This is essential

to solve the rotation ambiguity problem that arises when optimizing multiple principal directions simultaneously.

The variances along each direction are learned in an explicit manner, by encoding them in the norms of the unnormalized directions. This leads to the following loss term:

$$\mathcal{L}_\sigma = \sum_{(\mathbf{x}_m, \mathbf{x}, \hat{x}) \in \mathcal{D}'} \sum_{k=1}^K \left(\left\| \mathbf{d}_k - \sum_{l=1}^{k-1} (\mathbf{d}_k^T \mathbf{w}_l) \mathbf{w}_l \right\|^2 - \|[\mathbf{w}_k]_{\text{sg}}^T \mathbf{e}\|^2 \right). \quad (5)$$

Therefore, the overall loss function is given by:

$$\mathcal{L} = \mathcal{L}_w + \lambda \mathcal{L}_\sigma \quad (6)$$

where λ is a weight factor.

4. Experimental Setup

Dataset. We conducted our experiments over the LibriSpeech dataset, using train-clean-360 for training and test-clean for validation at a sampling rate of 16 kHz. The inpainting task was designed to reconstruct masked audio segments with a pre-determined duration of 128 ms, while the total input audio duration was set to 2 s during training. To ensure meaningful masking, we applied voice activity detection (VAD) to locate speech regions and randomly positioned 128 ms masks within these regions. This approach ensures that the model learns to inpaint speech segments rather than silence or non-speech regions. We performed all experiments on the normalized log-magnitude spectrogram computed with a window size of 256 and a hop length of 128. Various methods exist for phase reconstruction based on the estimated magnitude [21, 22]. However,

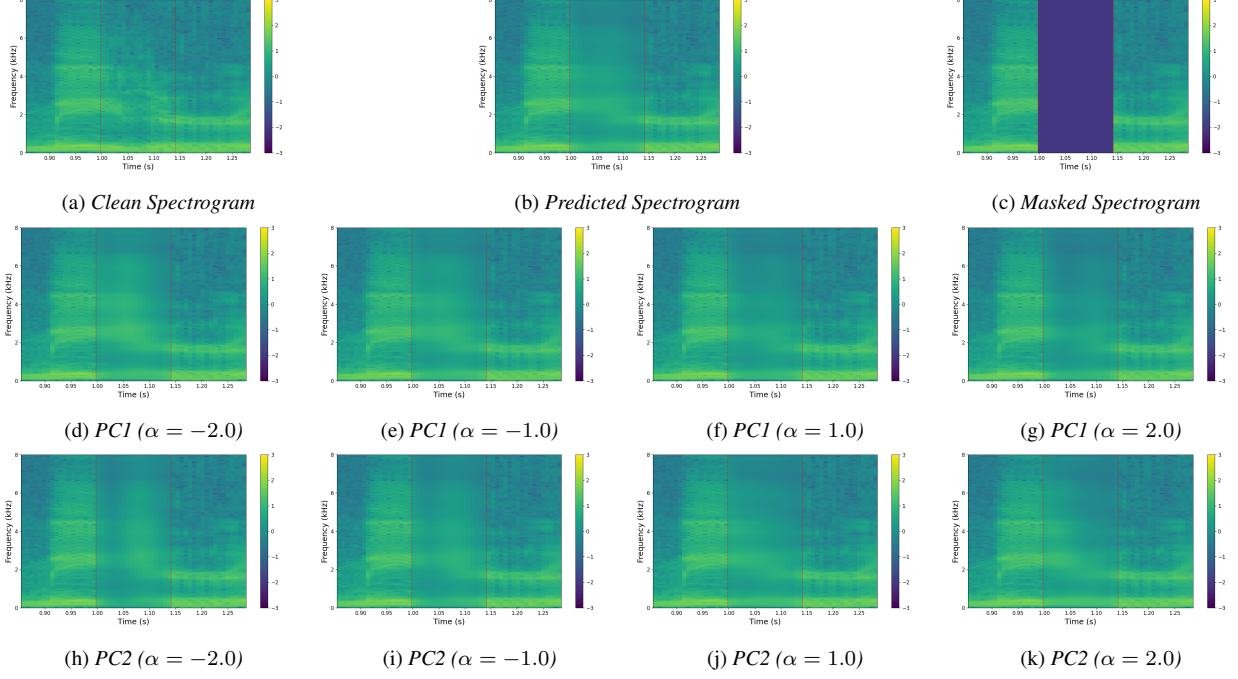


Figure 2: *Sample 3. Comparison of clean, masked, and PC projections for different α values. First row presents clean and masked spectrograms. Second and Third row correspond to variations along the first and second PCs, respectively.*

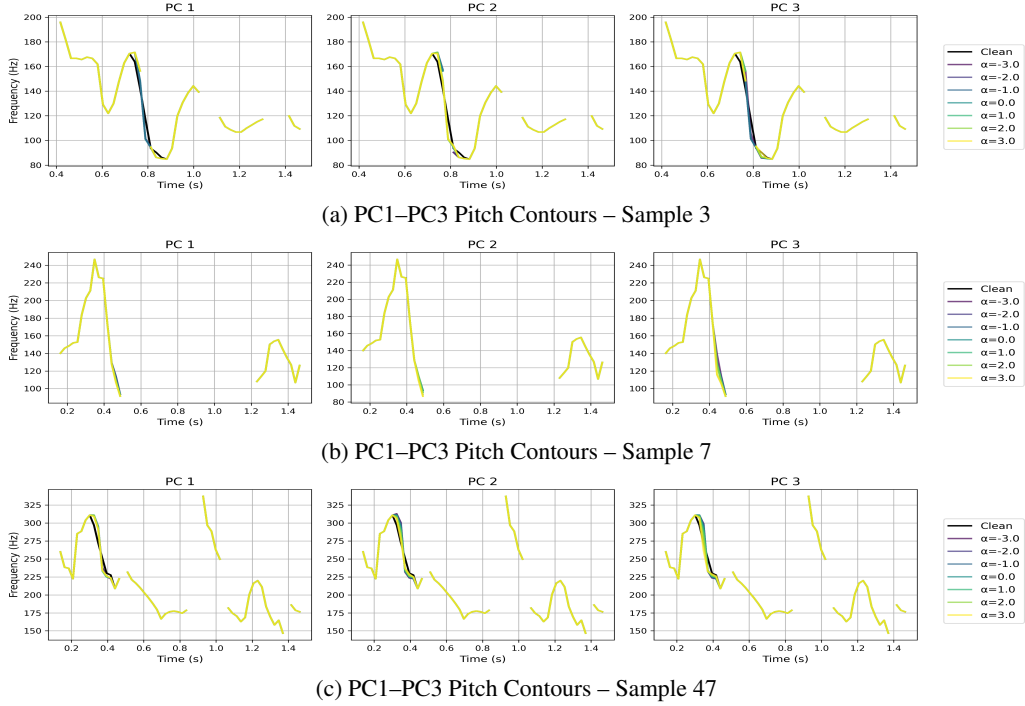


Figure 3: *Pitch contour variations across PC1-PC3 for samples 3, 7, and 47. Shared legend is shown on the right in each row.*

since this is not the focus of this paper, we use the phase of the clean signal directly to convert the spectrogram back to the time domain. Using the clean signal phase may slightly overestimate perceptual quality compared to blind reconstruction.

Model Architecture and Training Settings. Both the inpainting and PC models employ a U-Net with 8 blocks. The models have approximately 7.8M parameters, and their blocks and hyperparameters are summarized in Table 1. The NPPC

model outputs $K = 5$ PCs. Although more advanced U-Net variants or alternative architectures could be employed, these enhancements are independent of the uncertainty quantification framework, which is the primary focus of this paper.

Both the restoration model and the NPPC model are non-causal and were trained for 16 epochs using the Adam optimizer with a learning rate of $2e - 4$, $\beta_1 = 0.5$ and $\beta_2 = 0.999$. The weighting factor λ gradually increases from 0 to 1 over the first

Table 2: Transcriptions of the clean, masked, and output + α PCs signals. Differences from the clean transcription are highlighted.

Signal	α	Sample 3	Sample 47	Sample 7
Clean		We had meters in which there were two bottles of liquid	The Vanderpools were sure of this and the englishman	The paris plant, like that of the crystal palace, was a temporary exhibit
Masked		We had meters in which there were two bottles of liquid	The bandit pools were sure of this and the englishman	The parrots plant, like that of the crystal palace, was a temporary exhibit
PC1	-3.0	We had measures in which there were two bottles of liquid	The venture pools were sure of this and the englishman	The parrots plant, like that of the crystal palace, was a temporary exhibit
PC1	-2.0	We had mechers in which there were two bottles of liquid	The banter pools were sure of this and the englishman	The parrots plant, like that of the crystal palace, was a temporary exhibit
PC1	-1.0	We had mirrors in which there were two bottles of liquid	The banter pools were sure of this and the englishman	The parrots plant, like that of the crystal palace, was a temporary exhibit
PC1	0.0	We had meters in which there were two bottles of liquid	The banter pools were sure of this and the englishman	The parents plant, like that of the crystal palace, was a temporary exhibit
PC1	1.0	We had meters in which there were two bottles of liquid	The ventry pools were sure of this and the englishman	The parents plant, like that of the crystal palace, was a temporary exhibit
PC1	2.0	We had meters in which there were two bottles of liquid	The ventry pools were sure of this and the englishman	The parent's plant, like that of the crystal palace, was a temporary exhibit
PC1	3.0	We had meters in which there were two bottles of liquid	The vent repuls were sure of this and the englishman	The parent's plant, like that of the crystal palace, was a temporary exhibit
PC2	-3.0	We had meekers in which there were two bottles of liquid	The venture pools were sure of this and the englishman	The parents plant, like that of the crystal palace, was a temporary exhibit
PC2	-2.0	We had meekers in which there were two bottles of liquid	The ventipools were sure of this and the englishman	The parents plant, like that of the crystal palace, was a temporary exhibit
PC2	-1.0	We had meekers in which there were two bottles of liquid	The bento pools were sure of this and the englishman	The parrots plant, like that of the crystal palace, was a temporary exhibit
PC2	0.0	We had meters in which there were two bottles of liquid	The banter pools were sure of this and the englishman	The parents plant, like that of the crystal palace, was a temporary exhibit
PC2	1.0	We had mirrors in which there were two bottles of liquid	The bandit pools were sure of this and the englishman	The parent's plant, like that of the crystal palace, was a temporary exhibit
PC2	2.0	We had mirrors in which there were two bottles of liquid	The band repools were sure of this and the englishman	The parent's plant, like that of the crystal palace, was a temporary exhibit
PC2	3.0	We had mirrors in which there were two bottles of liquid	The band repools were sure of this and the englishman	The parent's plant, like that of the crystal palace, was a temporary exhibit
PC3	-3.0	We had meekers in which there were two bottles of liquid	The banter pools were sure of this and the englishman	The parrots plant, like that of the crystal palace, was a temporary exhibit
PC3	-2.0	We had meters in which there were two bottles of liquid	The banter pools were sure of this and the englishman	The parrots plant, like that of the crystal palace, was a temporary exhibit
PC3	-1.0	We had meters in which there were two bottles of liquid	The bandit pools were sure of this and the englishman	The parrots plant, like that of the crystal palace, was a temporary exhibit
PC3	0.0	We had meters in which there were two bottles of liquid	The banter pools were sure of this and the englishman	The parents plant, like that of the crystal palace, was a temporary exhibit
PC3	1.0	We had mirrors in which there were two bottles of liquid	The ventric pools were sure of this and the englishman	The parent's plant, like that of the crystal palace, was a temporary exhibit
PC3	2.0	We had mirrors in which there were two bottles of liquid	The ventric pools were sure of this and the englishman	The paris plant, like that of the crystal palace, was a temporary exhibit
PC3	3.0	We had mirrors in which there were two bottles of liquid	The vendor pools were sure of this and the englishman	The paris plant, like that of the crystal palace, was a temporary exhibit

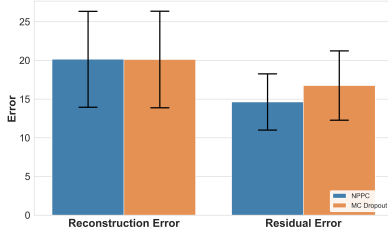


Figure 4: Comparison to MC Dropout in terms of RMSE and residual error. Error bars indicate the standard deviation.

500 training steps and remains fixed at 1 afterward.

Evaluation. For the transcription task, we use the Whisper model [23]. Pitch estimation is performed using probabilistic YIN (pYIN) [24], as implemented in the Librosa library. Additionally, we report the Root Mean Squared Error (RMSE) $\|\mathbf{x} - \hat{\mathbf{x}}\|^2$, and the residual error magnitude $\|\mathbf{e} - \mathbf{W}\mathbf{W}^T\mathbf{e}\|^2$, both computed in the time-frequency domain using only the magnitude of the spectrogram.

5. Results

We examine the differences among the PCs in terms of the obtained spectrograms as well as the variations in transcription and pitch. Additionally, we compare NPPC to a Bayesian deep learning approach in terms of the output Root Mean Squared Error (RMSE) and the reconstruction of the error.

Example Spectrograms. Fig. 2 displays the spectrograms of the original speech signal, the masked signal, and various transformations along the PCs. Given the small gap, we zoom in on the surrounding interval for clarity. The results indicate that different directions lead to distinct spectrogram variations, emphasizing different frequency regions. Our GitHub repository provides corresponding audio samples, enabling listeners to directly perceive the variations in speech content.

Transcription. Table 2 presents transcription examples for different traversal directions. The results show a variety of possible transcripts depending on both the direction and the traversal distance. For instance, the word “meters” is sometimes transcribed as “measures,” “meekers,” or “mirrors,”

while “Paris” appears as “parents” or “parrots”. Notably, in some cases, the initial output is incorrect, but moving along the PCs recovers the correct transcription.

Pitch. Fig. 3 compares the estimated pitch when moving across different PCs directions. While variations in pitch are observed along different PCs, they are generally less pronounced than content variations. This may be because pitch is largely speaker-dependent and can often be inferred with high confidence from the unmasked regions of the input signal.

RMSE and Residual Error. As a baseline for a quantitative comparison, we use MC Dropout [11]. MC Dropout is a technique used in Bayesian deep learning to estimate model uncertainty by enabling dropout at both training and inference time. During inference, the model performs multiple forward passes with different dropout masks, generating a distribution of predictions instead of a single deterministic output. We apply MC dropout to generate 50 output samples and perform principal component analysis (PCA) on this set, extracting the top 5 PCs.

The RMSE and residual error are shown in Fig. 4, averaged across 50 test samples. We observe that both approaches obtain comparable RMSE, while NPPC achieves slightly better reconstruction error. This is while being $50\times$ faster during inference than MC Dropout, since NPPC requires a single forward pass, while MC Dropout requires 50 passes.

6. Conclusions

In this paper, we emphasize the importance of modeling uncertainty in speech inpainting, moving beyond traditional approaches that generate a single deterministic output. We adopt the NPPC method, which learns to output the principal components of the posterior distribution. Our results demonstrate that NPPC better captures inpainting errors than the computationally intensive MC Dropout approach, while also producing perceptually diverse outputs, as illustrated in the accompanying audio samples. This can enable future systems to reflect ambiguity and adapt outputs to user intent.

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